AI-Powered Design of Organic Photovoltaic

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Motivation

Organic photovoltaics (OPV) depend on precise donor–acceptor design, where small molecular changes can significantly affect performance. Traditional discovery methods are slow and limited by the vast chemical space, motivating the need for AI-driven approaches to generate novel, high-performance molecules.

Challenge

- Huge chemical search space (millions of molecules).
- Limited labeled data for key properties.
- Need:
  - Chemical validity
  - Synthesizability
  - High performance (PCE)

Hypothesis

Combining graph-based molecular representations with semantic embeddings and generative modeling can better capture chemical relationships. A hybrid approach is expected to improve the generation of valid, synthesizable, and high-performance OPV candidates.

Solution: Hybrid Generative Pipeline

We propose a hybrid generative framework that integrates structural, semantic, and synthesis-aware information. Molecular graphs and embeddings are combined and processed through a variational graph autoencoder (VGAE) to generate novel donor–acceptor pairs optimized for performance and chemical feasibility.

System Pipeline

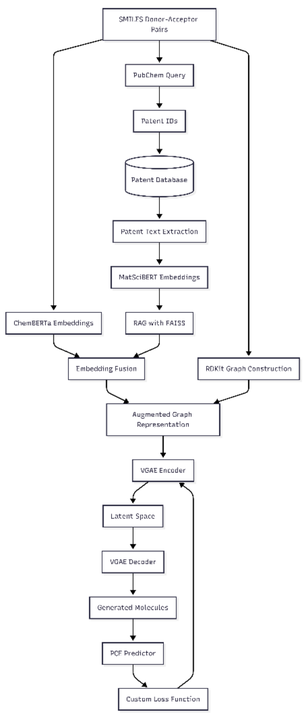

Figure 1: End-to-End OPV Generation Pipeline

SMILES representations are enriched with structural, synthesis, and retrieval features, then encoded via a VGAE into a molecular latent space. New candidates are sampled and evaluated for validity and predicted performance.

Model Architecture

The model integrates structural, semantic, and synthesis-aware information into a unified representation. Molecular graphs and embeddings are fused and processed through a variational graph autoencoder (VGAE), which learns a latent distribution of valid molecules. New candidates are generated from this space and evaluated using performance prediction and chemical constraints.

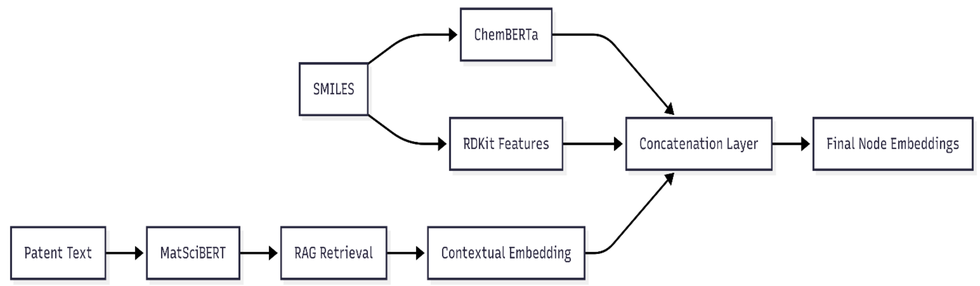

Figure 2: Multi-Embedding Fusion for Molecular Representation

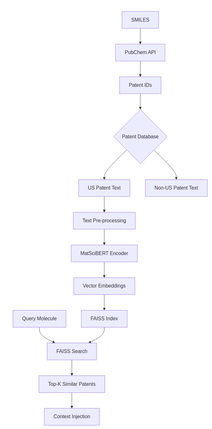

Figure 3: Data Processing and Embedding Pipeline

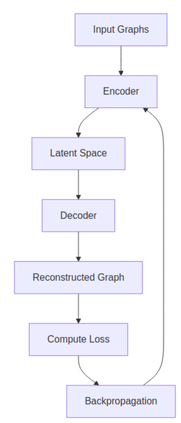

Figure 4: Variational Graph Autoencoder (VGAE)

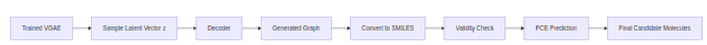

Figure 5: Molecule Generation and Evaluation Pipeline

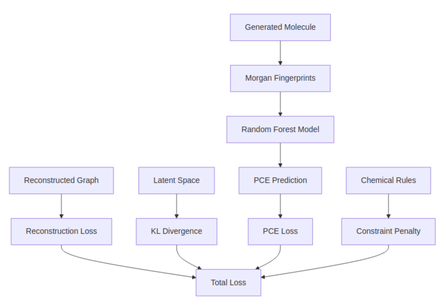

Figure 6: Multi-Objective Loss Function

Predictions

Expected model outputs
- Novel donor–acceptor SMILES pairs (VGAE)
- PCE score per pair (Random Forest)
- Chemical validity per molecule

Predictive metrics
- MAE / MSE • ROC-AUC • F1 • ELBO

Generative quality metrics
- MMD (WL Kernel) • KS Distance
- Degree Distribution • Clustering Coefficient

Evaluation targets
- High validity rate
- Competitive PCE vs. OPV benchmarks
- Structural diversity in generated pairs

Metrics

Custom multi-objective loss function

- Reconstruction Loss: graph fidelity
- KL Divergence: latent space regularization
- PCE Loss: performance optimization
- Constraint Penalty: chemical validity
- Total Loss: joint objective across all components

Evaluation metrics

- MAE / MSE • ROC-AUC • ELBO
- MMD (WL Kernel) • KS Distance • Clustering Coeff.

PCE Prediction Model
- Morgan Fingerprints → Random Forest → PCE Score

Limitations

- Investigate exploding and vanishing gradients for recent prediction period
- Investigate directional accuracy (still not stable to rely on as a practical option)

Future Work

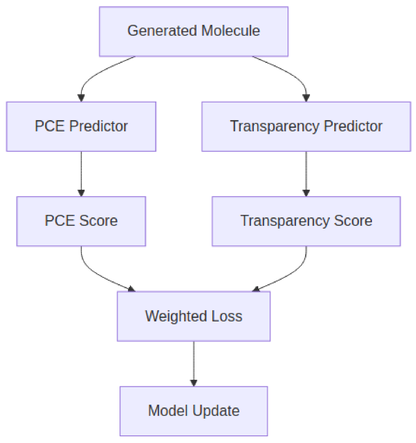

Figure 7: Generated Molecule Evaluation

Future work aims to close the generation–evaluation loop. Generated molecules will be assessed by dual predictors for power conversion efficiency (PCE) and molecular transparency. These scores feed a weighted loss function that drives iterative model updates, enabling self-improving optimization of donor–acceptor candidates over successive training cycles.

References

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